Sprint 2 – Search for Sprites

Student Information

Integrity Policy: All university integrity and class syllabus policies have been followed. I have neither given, nor received, nor have I tolerated others' use of unauthorized aid.

I understand and followed these policies: ◯ Yes ◯ No

Name:

Date:

Submission Details

Final ***Changelist*** number:

Verified build: ◯ Yes ◯ No

Required Configurations: DEBUG only

YouTubeLink:

Discussion (What did you learn):

Verify Builds

- Follow the Piazza procedure on submission
  - Verify your submission compiles and works at the changelist number.
- Verify that only MINIMUM files are submitted
  - No – Generated files
    - *.pdb, *.suo, *.sdf, *.user, *.obj, *.exe, *.log, *.pdb, *.db, *.user
    - Anything that is generated by the compiler should not be included
  - No – Generated directories
    - /Debug, /Release, /Log, /ipch, /.vs
- Typical files project files that are required
  - *.sln, *.csproj, *.cs,
  - App.config, AssemblyInfo.cs, CleanMe.bat
  - Resources Directory:
    - *.tga, *.dll, *.wav, *.glsl, *.azul

Standard Rules

Submit multiple times to Perforce

- Submit your work as you go to perforce several times
  - Design Patterns – no minimum
  - Sprints – minimum of 5 real submissions (generally 10+)
    - As soon as you get something working, submit to perforce
    - Have reasonable check-in comments
      - Points will be deducted if minimum is not reached

Submission Report

- Fill out the submission Report
  - No report, no grade

Code and project needs to compile and run

- Make sure that your program compiles and runs
  - Warning level 4
  - NO Warnings or ERRORS
    - Your code should be squeaky clean.
  - Code needs to work “as-is”.
    - No modifications to files or deleting files necessary to compile or run.
  - All your code must compile from perforce with no modifications.
    - Otherwise it's a 0, no exceptions

Project needs to run to completion

- If it crashes for any reason…
  - It will not be graded and you get a 0

No Containers

- Containers (No automatic containers or arrays
- Template or generic parameters
- No arrays
  - You need to do this the old fashion way - ***YOU EARNED IT***

Leave Project Settings

- Do NOT change the project or warning level
  - Any changing of level or suppression of warnings is an integrity issue

Simple C#

- No .Net
- We are using the basics
  - Types:
    - Class, Structs, intrinsic types (int, float, bool, etc…)
    - NO arrays allowed!
  - Basics language features
    - Inheritance, methods, abstract, virtual, etc…

No Debug code or files disabled

- Make sure the program has only active code
  - If you added debug code or commented out code,
    - please return to code to active state or remove it

Adding files to this project

- Make sure you add the files in the appropriate sub-directories
- Make sure any new files are successfully integrated into the project
- Make sure your new files are submitted to Perforce

Due Dates

- See Piazza for due date and time
- Submit program perforce in your student directory assignment supplied.
- Fill out your this ***Submission Report*** and commit to perforce
  - ***ONLY*** use Adobe Reader to fill out form, all others will be rejected.
  - Fill out the form and discussion for full credit.

Goals

- Sprite – abstraction
  - Image
  - Texture
- Refactor
  - Learn and understand the code
  - Adding singletons
  - Refactoring to use SLink
- Understand Texture/Image layout
  - Create several new texture/images

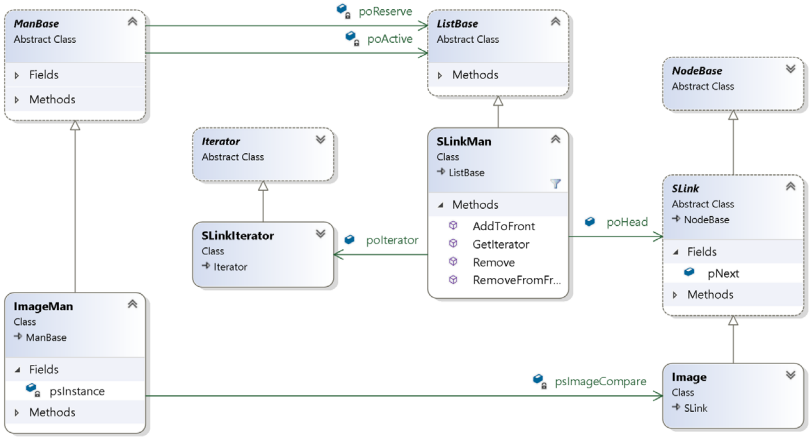
Assignments

Grading

- This sprint you are going to learn/modify the code from class.
- There are 5 steps to this refactoring

Step 1 – Refactoring 2 managers

- Refactor **TextMan** to use Slink
  - Fix the manager to use SLink
  - Since the textures are never going to removed
  - Designed to be loaded and consistently reused in the game
  - No benefit to DLink… when SLink will work fine.
- Refactor **ImageMan** to use Slink
  - Fix the manager to use Slink



Step 2 – Add Singleton to SpriteMan

- Refactor SpriteMan to use singleton
  - Use the ImageMan as a reference
  - `SpriteMan.Create(2, 1);`
  - `SpriteMan.Destroy();`

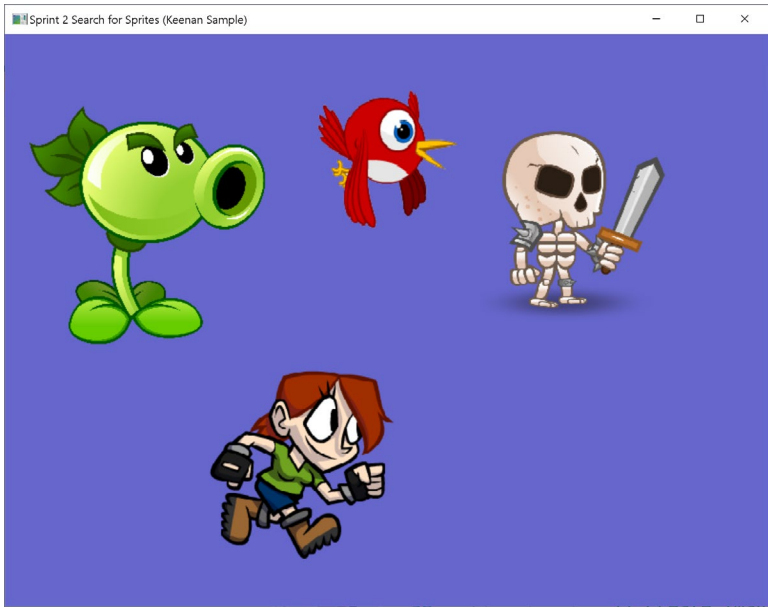
Step 3 – Remove the news in LoadContent()

- There should be no news in the LoadContent() method
  - Since pass the individual data elements directly
- Since the managers are now all Singletons there is no visible news in LoadContent()
  - ```
    SpriteMan.Add(Sprite.Name.RedBird,
                  pImageRedBird,
                  new Azul.Rect(50, 500, 50, 50));
    ```

Step 4 – Remove the variables in LoadContent() and globals from Game class

- LoadContent()
  - There should be no "variables" in the LoadContent() method
  - This forces the managers to Find() the resource
  - Local variables from LoadContent() such as:
    - `Image pImageRedBird`
    - `Image pImageYellowBird`
    - `Image pImageGreenBird`
    - `Image pImageWhiteBird`
- Game class
  - Remove these variables from the Game class
    - `Sprite pRedBird;`
    - `Sprite pWhiteBird;`
    - `Sprite pYellowBird;`
    - `Sprite pGreenBird;`
  - Rework the LoadContent(), Update(), Draw() to use the manager's Find();
    ```

Step 5 – Mimic the following sample demo:



- Here is a video showing the movement:
  - https://youtu.be/UblDLgOqExY
  - There is motion in the video.. mimic that as well
- This demo has 4 textures with 4 images
  - Use some tool like Gimp or photo shop to get the Image Rectangles
    - PeaShooter2.tga
    - Skeleton2.tga
    - Running2.tga
    - Flying2.tga
  - Use the converted texture to load into the game
    - PeaShooter2.t.azul
    - Skeleton2.t.azul
    - Running2.t.azul
    - Flying2.t.azul
- Use a tool to measure the Rectangle Box inside the texture
  - I suggest Gimp (free) start a thread on how to find the box

Required Features

Features to show off in Demo

Show it off and talk about these required features

- Game playing mimic the sample demo
- Show the Game.cs – show off LoadContent()
- Should see the manager being used
- No visible news in LoadContent()
- Show the UML diagram of ImageMan()

Video Capture

- Have Visual Studio open and ready to go.
- Start recording
    - Show output window running
    - Speak clearly and loud
    - Discuss the features you added in this demo
    - Duration
        - Approx. 2-3 min long YouTube video capture
            - Do not exceed 5 min
        - Use OBS or similar for recording
            - High resolution / good framerate
            - Good sound quality
            - Show on the capture… it compiling and then running
        - Place YouTube link in PDF
        - Make sure it public and runs without passwords or login
        - Watch your video
            - Can you see the code clearly
            - Can you hear your voice clearly
- Don't lose points
    - Make sure everything is submitted properly
    - Don't record beyond 5 min.
    - You audio volume was too quiet (test it and verify it)
    - Your YouTube link needs to work "As-IS" with no login
        - Don't make it private… make it unlisted
    - You check-in too many files or temporary files
        - Make sure you run CleanMe.bat before perforce submission

General guidelines:

- Post on Piazza questions and clarifications
- Make sure you delete these using directives (we are not using them)
    - `using System.Collections.Generic;`
    - `using System.Linq;`
    - `using System.Text;`
    - `using System.Threading.Tasks;`

Development

- Use the project in Sprint2
- Do not copy temporary files or submit extra data
- Make sure your run CleanMe.bat before adding to perforce

Submission

- Submit your SprintX directory into perforce:
  - /student/<yourname>/SprintX/...
    - You need to submit a complete C# project
  - Solution, project and C# files (whatever it takes to build the project)
    - Do not submit anything that is auto generated
  - Run the supplied CleanMe.bat before submission
    - Should cleanup files
- Fill out the Submission report and submit that pdf to your student directory

Validation

Simple checklist to make sure that everything is submitted correctly

- Is the project compiling and running without any errors or warnings?
- Does the project runs ***ALL*** without crashing?
- Is the submission report filled in and submitted to perforce?
- Follow the verification process for perforce
  - Is all the code there and compiles “as-is”?
  - No extra files
- Submit the YouTube link to PDF

Hints

Most assignments will have hints in a section like this.

- Watch the lecture… look at the code supplied in class
- Study the projects in class lecture

Troubleshooting

- Print to the output window to help you debug
  - It really helps
- Ask for clarification on piazza